

Structure Growth in the Gravitating Vacuum Model:

The Void–Cluster Cell, Modified Perturbation Equations, and the S_8 Tension

Boris Kriger^{1,2}

¹ Information Physics Institute, Gosport
boris.kriger@informationphysicsinstitute.net

² Institute of Integrative and Interdisciplinary Research, Toronto
boriskriger@interdisciplinary-institute.org

ORCID: [0009-0001-0034-2903](https://orcid.org/0009-0001-0034-2903)

Abstract

The gravitating vacuum model proposes that quantum vacuum energy and the cosmological constant are physically distinct: Λ is a geometric property of spacetime driving homogeneous expansion, while vacuum energy ρ_{vac} is a local field quantity that gravitates according to $E = mc^2$ and general relativity, with bare vacuum evolving as $(1+z)^3$ (a self-consistency requirement, not a free parameter). This paper develops the first quantitative treatment of structure growth within this framework. We construct a self-consistent *void–cluster cell*—a spherical void containing unsuppressed vacuum energy, bounded by a shell of matter hosting a gravitationally bound cluster where vacuum energy is suppressed to $\Lambda_0 - \alpha \rho_{\text{m}}$. We derive the modified linear perturbation equations including a volume-weighted Poisson source factor and solve the modified growth equation numerically.

The central result is a **duality between two observational tensions**: the sign of the vacuum–matter coupling α determines which tension is alleviated. For $\alpha > 0$ (vacuum suppressed by matter), growth is enhanced at all epochs—naturally explaining the massive high-redshift galaxies observed by JWST ($D/D_{\Lambda\text{CDM}} \approx 1 + 3\alpha$ at $z = 10$), but *worsening* the S_8 tension. For $\alpha < 0$ (vacuum enhanced by matter, the sign favored by running vacuum model fits with $\nu > 0$), growth is suppressed—naturally alleviating the S_8 tension ($\Delta S_8/S_8 \approx 3\alpha$) but making the JWST puzzle harder. We argue that both signs may operate at different epochs or density regimes, a possibility supported by the microscopic analysis of the vacuum–matter coupling.

Additional predictions include: enhanced ISW signal from voids, cumulative gravitational redshift for photons traversing de Sitter void interiors, and qualitatively different clock rates in voids compared to both ΛCDM and timescape cosmology. The de Sitter junction conditions at the void–wall boundary are derived, connecting the interior vacuum geometry to the cosmic web structure.

Keywords: vacuum energy, structure growth, S_8 tension, cosmic voids, perturbation theory, dark energy, dark matter, growth factor, integrated Sachs–Wolfe effect, void–cluster model

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1 Introduction

The standard Λ CDM cosmological model accounts for the cosmic microwave background (CMB) anisotropies (Planck Collaboration, 2020), the accelerating expansion inferred from Type Ia supernovae (Riess et al., 1998; Perlmutter et al., 1999), baryon acoustic oscillations (BAO) (Eisenstein et al., 2005; DESI Collaboration, 2024), and the large-scale distribution of galaxies. Yet several persistent tensions have emerged that suggest the need for deeper physical interpretation.

The most relevant for this paper is the S_8 tension. The parameter $S_8 \equiv \sigma_8(\Omega_m/0.3)^{0.5}$, which measures the amplitude of matter fluctuations weighted by the matter density, shows a systematic discrepancy between early-universe and late-universe measurements. The Planck CMB analysis yields $S_8 = 0.832 \pm 0.013$ (Planck Collaboration, 2020), while weak gravitational lensing surveys consistently find lower values: KiDS-1000 reports $S_8 = 0.759^{+0.024}_{-0.021}$ (Asgari et al., 2021), DES-Y3 gives $S_8 = 0.776 \pm 0.017$ (DES Collaboration, 2022), and HSC-Y3 finds $S_8 = 0.769^{+0.031}_{-0.034}$ (Li et al., 2023). The discrepancy stands at approximately $2\text{--}3\sigma$ and has persisted across multiple independent analyses.

This tension has a simple physical interpretation: the late-time universe is *less clumpy* than the early-universe calibration predicts. Either the growth of structure is suppressed at late times relative to Λ CDM, or the initial amplitude is overestimated. The former possibility is particularly interesting because it connects to the physics of cosmic voids, dark energy, and the nature of vacuum energy.

This paper develops a quantitative treatment of structure growth within the *gravitating vacuum* framework (Kriger, 2026a,b,c). The central hypothesis of this framework is that the quantum vacuum energy density ρ_{vac} and the cosmological constant Λ are physically distinct quantities. Λ is a geometric property of spacetime—constant, universal, driving homogeneous expansion. Vacuum energy is a local field quantity that gravitates according to $E = mc^2$ and general relativity, depends on the local matter density through the ansatz

$$\rho_{\text{vac}}(\rho_m) = \Lambda_0 - \alpha \rho_m, \quad (1)$$

and exhibits different dynamical behavior depending on whether it resides in an expanding (void) or bound (cluster) phase.

The key physical insight is that voids are not empty—they contain the highest vacuum energy density and, through the de Sitter equation of state $w = -1$, actively accelerate their own expansion and expel matter toward their boundaries. Clusters and filaments, conversely, are regions where vacuum energy is suppressed by the presence of matter, expansion is absent, and the residual vacuum gravitates as an effective pressureless component ($w_{\text{eff}} \approx 0$). The boundary between these phases—the turnaround radius—is a physical phase transition, not an arbitrary cutoff.

We construct the minimal self-consistent unit of this model: the *void-cluster cell*, consisting of a single spherical void surrounded by a matter shell hosting a bound cluster. We derive the perturbation equations, solve the modified growth equation, and show that the model naturally produces a lower late-time σ_8 —in the direction required to alleviate the S_8 tension.

The paper is organized as follows. Section 2 presents the logical structure of the argument. Section 3 reviews the relevant observational and theoretical background. Section 4 develops

the conceptual framework of the void–cluster cell. Section 5 contains the mathematical development, including modified perturbation equations and numerical solutions. Section 6 provides empirical illustrations. Section 7 discusses the results and compares with alternative models. Sections 8 and 9 address limitations and future directions.

2 Logical Structure of the Argument

The argument of this paper proceeds through the following sequence of steps.

Step 1 (§3): We review the observational evidence for the S_8 tension and the physics of cosmic voids, establishing that both phenomena require explanation and that they may be connected.

Step 2 (§4): We construct the void–cluster cell—the minimal self-consistent unit of the gravitating vacuum model. The key physical content is the identification of two vacuum phases: the *expanding phase* (void interior, $w = -1$, vacuum acts as repulsive dark energy) and the *bound phase* (cluster interior, $w_{\text{eff}} \approx 0$, vacuum gravitates as pressureless matter). The phase boundary is identified with the turnaround radius.

Step 3 (§5.2): We derive the dynamical equations for the void interior, showing that the de Sitter geometry produces faster-than- Λ CDM expansion, enhanced matter expulsion, and cumulative gravitational redshift for traversing photons.

Step 4 (§5.4): We derive the effective gravitational dynamics in the bound phase, showing that the suppressed vacuum energy produces $G_{\text{eff}} = G(1+2\alpha)$ and that the effective equation of state transitions from $w = -1$ to $w_{\text{eff}} \approx 0$.

Step 5 (§5.5): We derive the modified linear perturbation equations for the two-phase medium, including the modified growth equation for the linear growth factor $D(a)$.

Step 6 (§5.9): We solve the modified growth equation numerically for several values of α and compute $f\sigma_8(z)$ and S_8 .

Step 7 (§6): We compare the predictions with observational data and show that the model naturally alleviates the S_8 tension.

Synthesis: The void–cluster cell, governed by a single parameter α , produces a volume-weighted suppression of late-time growth that is consistent with the observed S_8 discrepancy while preserving CMB compatibility at early times.

3 Background and Related Work

3.1 The S_8 tension

The S_8 parameter combines the matter fluctuation amplitude σ_8 with the matter density Ω_m in the combination $S_8 = \sigma_8(\Omega_m/0.3)^{0.5}$ that is well-constrained by weak gravitational lensing. The Planck satellite, measuring the CMB at $z \approx 1100$ and evolving to the present using Λ CDM growth predictions, obtains $S_8 = 0.832 \pm 0.013$ (Planck Collaboration, 2020). Multiple late-time probes find systematically lower values: KiDS-1000 (Asgari et al., 2021), DES-Y3 (DES Collaboration, 2022), HSC-Y3 (Li et al., 2023), and combined analyses (Di Valentino et al., 2021) all cluster around $S_8 \approx 0.76$ – 0.78 , a 2 – 3σ deficit.

This discrepancy could arise from: (a) systematic errors in either CMB or lensing analyses, (b) new physics that suppresses late-time growth, or (c) a modification of the relationship between early and late amplitudes. The running vacuum model (RVM) of Solà Peracaula and collaborators (Solà, 2013; Solà Peracaula et al., 2017, 2021) has shown that a mildly dynamical vacuum ($\rho_{\text{vac}} \propto H^2$) can alleviate both S_8 and H_0 tensions at the 2–3 σ level. The gravitating vacuum model is structurally related to the RVM through the Friedmann equation (see Section 5.10).

3.2 Cosmic void observations

Cosmic voids occupy approximately 60% of the cosmic volume (Pan et al., 2012; Hamaus et al., 2014). Stacked void profiles from the SDSS and BOSS surveys reveal a universal density profile: a central underdensity $\delta \approx -0.8$ to -0.9 , a compensating ridge at the void boundary, and a return to the cosmic mean at ~ 2 void radii (Hamaus et al., 2014; Nadathur et al., 2015). Peculiar velocities around voids, measured through redshift-space distortions, show systematic outflow from void centers (Hamaus et al., 2017; Achitouv et al., 2017).

The integrated Sachs–Wolfe (ISW) effect—the net energy shift of CMB photons traversing time-evolving gravitational potentials—provides a direct probe of the void interior. Granett et al. (2008) reported an ISW signal from SDSS superstructures that is 2–3 times stronger than the Λ CDM prediction. While subsequent analyses have debated the significance (Hernández-Monteagudo & Smith, 2013; Cai et al., 2014; Nadathur & Crittenden, 2016), the direction of the anomaly (excess signal from voids) is consistent with the prediction of the gravitating vacuum model.

3.3 Structure formation in Λ CDM

In Λ CDM, the linear growth factor $D(a)$ satisfies

$$D'' + \left(2 + \frac{H'}{H}\right) D' - \frac{3}{2} \Omega_{\text{m}}(a) D = 0, \quad (2)$$

where primes denote $d/d \ln a$ and $\Omega_{\text{m}}(a) = \Omega_{\text{m},0} a^{-3}/[H(a)/H_0]^2$. The growth rate is parameterized as $f = d \ln D / d \ln a \approx \Omega_{\text{m}}(a)^{0.55}$ (Linder, 2005), and the observable $f\sigma_8(z)$ is measured through redshift-space distortions (RSD) (Percival et al., 2009; Blake et al., 2011; Beutler et al., 2012; Alam et al., 2017).

The Λ CDM prediction for $f\sigma_8(z)$ is fixed once the CMB normalization σ_8 and Ω_{m} are specified. The S_8 tension states that this prediction is systematically too high at $z \lesssim 1$.

3.4 Counterarguments and Responses

Objection: The S_8 tension is only 2–3 σ and may be a statistical fluctuation or systematic error.

Response: The tension persists across multiple independent surveys (KiDS, DES, HSC) using different pipelines, data, and systematic treatments. While no single measurement is decisive, the coherent direction of the discrepancy across independent analyses makes a

physical explanation worth investigating. Our model provides a specific mechanism whose predictions can be tested.

Objection: The RVM already addresses the S_8 tension. What does the gravitating vacuum model add?

Response: The RVM describes a temporal variation of vacuum energy: $\rho_{\text{vac}}(H)$. The gravitating vacuum model describes a spatial variation at fixed epoch: $\rho_{\text{vac}}(\rho_{\text{m}})$. Through the Friedmann equation, the two are mathematically related (see Section 5.10), but the spatial picture provides new observational predictions (void profiles, ISW enhancement, Hubble diagram correlations with void fraction) that the purely temporal RVM does not make.

3.5 Section Result and Implications

The S_8 tension is a persistent observational discrepancy that requires the late-time growth of structure to be suppressed relative to Λ CDM. Cosmic voids—which occupy most of the cosmic volume—are the natural arena in which such suppression could originate. The gravitating vacuum model provides a specific physical mechanism: vacuum energy in voids suppresses growth, while enhanced gravity in bound structures partially compensates. The net volume-weighted effect is a reduction of S_8 .

4 Conceptual Framework: The Void–Cluster Cell

4.1 The minimal self-consistent unit

The cosmic web consists of voids, walls, filaments, and clusters. The minimal self-consistent unit that captures the essential physics of the gravitating vacuum model is the *void–cluster cell*: a single spherical void of radius R_v , surrounded by a shell of matter (wall) of thickness ΔR , hosting a gravitationally bound cluster at the void boundary.

This is a simplified geometry—real voids are not perfectly spherical, and the cosmic web is topologically complex. But the void–cluster cell captures the two essential phases (expanding and bound), the transition between them, and the volume-weighting that determines the effective growth rate.

4.2 Two phases of vacuum energy

The physical content of the model resides in the recognition that vacuum energy manifests differently depending on whether the local spacetime is expanding or bound.

Definition 4.1 (Expanding phase). In a region where the local Hubble rate $H_{\text{local}} > 0$ (cosmic expansion is active), vacuum energy has equation of state $w = -1$, negative pressure $p = -\rho_{\text{vac}} c^2$, and the effective gravitational source is

$$\rho_{\text{vac}} + 3p_{\text{vac}}/c^2 = \rho_{\text{vac}} - 3\rho_{\text{vac}} = -2\rho_{\text{vac}} < 0. \quad (3)$$

The vacuum drives accelerated expansion and repels matter.

Definition 4.2 (Bound phase). In a gravitationally bound region where $H_{\text{local}} = 0$ (expansion is suppressed), the negative pressure of the vacuum cannot perform work (there is no expansion for it to act on). The dynamically relevant quantity is only the mass-energy $E = mc^2$. The effective equation of state is $w_{\text{eff}} \approx 0$, and the vacuum gravitates as pressureless matter.

Remark 4.3 (Physical basis for the phase transition: de Sitter–Schwarzschild analysis). The claim that vacuum energy contributes differently in bound vs. expanding regions requires careful justification, since in GR the stress-energy tensor $T_{\mu\nu}$ sources gravity regardless of the local kinematics. We address this through the de Sitter–Schwarzschild (dS–S) metric, which describes a mass M embedded in a vacuum with energy density ρ_{vac} :

$$ds^2 = -\left(1 - \frac{2GM}{rc^2} - \frac{H_v^2 r^2}{c^2}\right) c^2 dt^2 + \dots \quad (4)$$

where $H_v = \sqrt{8\pi G \rho_{\text{vac}}/3}$. The gravitational acceleration has two terms: $-GM/r^2$ (attractive, from matter) and $+H_v^2 r$ (repulsive, from vacuum). The vacuum term dominates at the *vacuum radius*:

$$r_{\text{vac}} = \left(\frac{2GM}{H_v^2}\right)^{1/3}. \quad (5)$$

For a cluster of mass $M \sim 10^{14} M_{\odot}$ and $\rho_{\text{vac}} \sim 4.5 \times 10^{-24} \text{ kg m}^{-3}$: $r_{\text{vac}} \approx 0.7 \text{ Mpc}$, comparable to the virial radius. **Inside the bound region** ($r \ll r_{\text{vac}}$), the vacuum repulsion $H_v^2 r$ is suppressed relative to the gravitational attraction by a factor $(r/r_{\text{vac}})^3 \ll 1$. The vacuum contributes negligibly to local dynamics—not because its equation of state changes, but because its *dynamical effect is geometrically suppressed* inside the deep potential well.

We do *not* claim that w changes from -1 to 0 . We claim that the vacuum’s dynamical contribution to bound-system physics is suppressed by the geometric factor $(r/r_{\text{vac}})^3$. The Komar mass integral still includes $\rho + 3p/c^2 = -2\rho_{\text{vac}}$, but this integral over the small bound region yields a correction of order $(R_{\text{vir}}/r_{\text{vac}})^3 \sim \mathcal{O}(1)$ —significant but not dominant.

The earlier heuristic equation $w_{\text{eff}} = w \cdot (\dot{H}_{\text{local}}/\dot{H}_{\text{background}})$ is therefore **withdrawn** as insufficiently rigorous. The correct statement is: the vacuum’s contribution to local gravitational dynamics is geometrically suppressed in bound regions by the factor $(r/r_{\text{vac}})^3$, approaching full GR repulsion only at $r \gtrsim r_{\text{vac}}$ (the void boundary).

4.3 The void interior

Inside the void, matter density is negligible: $\rho_{\text{m}} \rightarrow 0$. The vacuum energy is unsuppressed:

$$\rho_{\text{vac}}^{(\text{void})} \approx \Lambda_0. \quad (6)$$

The geometry is approximately de Sitter. The static form of the de Sitter metric is

$$ds^2 = -\left(1 - \frac{H_v^2 r^2}{c^2}\right) c^2 dt^2 + \left(1 - \frac{H_v^2 r^2}{c^2}\right)^{-1} dr^2 + r^2 d\Omega^2, \quad (7)$$

where $H_v = \sqrt{8\pi G \Lambda_0/3}$ is the expansion rate inside the void.

The total vacuum mass-energy inside a void of radius $R_v \sim 30$ Mpc is

$$M_{\text{vac}} = \frac{4}{3}\pi R_v^3 \rho_{\text{vac}} \sim 10^{18} M_{\odot}, \quad (8)$$

if $\rho_{\text{vac}} \sim 4.5 \times 10^{-24} \text{ kg m}^{-3}$ (the value estimated in [Kriger 2026a](#) from galactic halo masses). This is a colossal energy, far exceeding any galaxy cluster. But it does not cause collapse—it causes repulsion, because $\rho_{\text{vac}} + 3p/c^2 = -2\rho_{\text{vac}} < 0$.

4.4 The bound cluster

Inside the cluster, matter density is high: $\rho_{\text{m}} \gg 0$. Vacuum energy is suppressed:

$$\rho_{\text{vac}}^{(\text{cluster})} = \Lambda_0 - \alpha \rho_{\text{m}}. \quad (9)$$

Expansion is absent ($H_{\text{local}} = 0$). The Poisson equation is

$$\nabla^2 \Phi = 4\pi G(1 + 2\alpha) \rho_{\text{m}} - \Lambda_0, \quad (10)$$

where the constant Λ_0 contributes only a uniform background with no local dynamical effect. The effective gravitational coupling is

$$G_{\text{eff}} = G(1 + 2\alpha). \quad (11)$$

4.5 Why galaxies do not fall into voids

The void is the most massive “object” in this model ($\sim 10^{18} M_{\odot}$). Why don’t galaxies fall into it?

Because in general relativity, gravity is sourced not by mass alone but by $\rho + 3p/c^2$. For vacuum with $w = -1$:

$$\rho_{\text{vac}} + 3p_{\text{vac}}/c^2 = -2\rho_{\text{vac}} < 0. \quad (12)$$

The effective source is *negative*. The void repels, not attracts. The Raychaudhuri equation confirms this: geodesics inside a de Sitter space *diverge*. Galaxies on the boundary of a void are pushed outward, toward the walls and filaments.

The void is not a gravitational well. It is a gravitational hill. Matter slides off it.

4.6 The void surface as a physical membrane

However—and this is the crucial subtlety raised in the discussion preceding this paper—a galaxy on the void boundary does not simply feel “repulsion.” It moves along a geodesic of the full spacetime metric, which is determined by the *total* energy content, including the colossal vacuum energy of the void.

The metric near the void boundary has large Christoffel symbols because the curvature is significant. A galaxy with a tangential velocity component (acquired during initial void formation or through tidal interactions) will follow a geodesic that curves along the void surface. The metric does not care whether the energy that curves it has $w = -1$ or $w = 0$ —curvature is curvature.

This means that the void surface acts as a *gravitational lens on megaparsec scales*. Galaxies are channeled along the void boundary toward the nearest node (cluster) via geodesics of the strongly curved spacetime. Their velocities are high because the curvature is large.

Proposition 4.4 (Tangential velocities on the void boundary). *In the gravitating vacuum model, the peculiar velocities of galaxies near void boundaries have a tangential component (along the void surface) that is enhanced relative to Λ CDM by the curvature contribution of the void’s vacuum energy. The enhancement scales as*

$$\frac{v_{\text{model}}}{v_{\Lambda\text{CDM}}} \approx \sqrt{1 + \frac{8\pi G \rho_{\text{vac}}}{3 H_0^2} \delta_v}, \quad (13)$$

where δ_v is the void density contrast.

This is a testable prediction: peculiar velocities near large voids should be systematically higher than Λ CDM predicts, particularly in the tangential direction.

4.7 Counterarguments and Responses

Objection: If the void contains $10^{18} M_\odot$ of vacuum energy, why doesn’t the universe collapse?

Response: Because vacuum energy has $w = -1$, and its effective gravitational source $\rho + 3p/c^2 = -2\rho$ is negative. More vacuum energy means more repulsion, not more attraction. The universe accelerates precisely because of the vacuum energy. The more of it there is, the faster it expands.

Objection: How can the same vacuum be repulsive in voids and attractive in clusters?

Response: The vacuum itself does not change. Its dynamical manifestation depends on whether the local spacetime is expanding. In an expanding void, the negative pressure $p = -\rho c^2$ performs work, driving acceleration. In a bound cluster, there is no expansion for the pressure to act on. What remains is only the mass-energy $E = mc^2$ —which gravitates attractively. The analogy is water under pressure: in a pipe with flow, pressure drives motion; in a sealed container, it does not.

Objection: The de Sitter metric inside a void requires $\rho_{\text{vac}} \sim 4.5 \times 10^{-24} \text{ kg m}^{-3}$, which is ~ 800 times the observed dark energy density. This is a new fine-tuning.

Response: This is a discrepancy of 10^3 , not 10^{120} . And the comparison is legitimate only if $\Lambda = 8\pi G \rho_{\text{vac}}$ —the very identification the gravitating vacuum model rejects. If Λ (geometry) and ρ_{vac} (field energy) are different quantities, there is no reason they should have the same value.

4.8 Section Result and Implications

The void–cluster cell provides a self-consistent minimal model with two vacuum phases: repulsive (de Sitter, $w = -1$) in the expanding void interior, and attractive ($w_{\text{eff}} \approx 0$) in the bound cluster. The transition occurs at the turnaround radius. Galaxies do not fall into voids because the vacuum there is repulsive. Instead, they are channeled along the void surface by geodesics of the curved spacetime, explaining the observed high peculiar velocities in the cosmic web.

5 Mathematical Development

5.1 Vacuum energy evolution: the $(1+z)^3$ scaling

A critical element of the model, derived in Kriger (2026a), is that the bare vacuum energy density evolves with cosmic expansion:

$$\Lambda_0(z) = \Lambda_{0,0} (1+z)^3, \quad (14)$$

where $\Lambda_{0,0}$ is the present-day value. This scaling is *not derived from a covariant action*; it is a motivated assumption whose consequences we explore. The motivation is twofold:

Algebraic motivation: The suppression term $\alpha\rho_m$ scales as $(1+z)^3$ (matter dilution). For the ansatz $\rho_{\text{vac}} = \Lambda_0 - \alpha\rho_m$ to remain well-defined at all epochs—specifically, for ρ_{vac} to remain positive and for α to remain constant—both terms must scale identically. This is a consistency condition on the ansatz, not a derivation from first principles.

Physical motivation (mode counting): The vacuum energy arises from the zero-point energy of quantum field modes. The number of available modes in a comoving volume scales with the physical volume, which for a given comoving region decreases as $a^{-3} = (1+z)^3$ as we look to higher redshift. If the vacuum energy density is set by the mode density (as in the Casimir effect, where modifying the mode spectrum changes the vacuum energy), then $\rho_{\text{vac}} \propto (\text{mode density}) \propto (1+z)^3$ is a natural scaling. This argument is heuristic—a rigorous derivation would require computing the one-loop effective action in FLRW spacetime with the full Standard Model field content, as initiated in Kriger (2026c)—but it provides a physical basis beyond mere algebraic consistency.

Consequences if the scaling fails: If Λ_0 scales as $(1+z)^n$ with $n \neq 3$, the ansatz breaks down: $n < 3$ drives ρ_{vac} negative at high z ; $n > 3$ makes vacuum dominate at high z . Only $n = 3$ preserves the structure. We regard this as a *falsifiable structural prediction* of the ansatz rather than a derived result.

The consequences are profound. At $z \approx 12$ (JWST epoch): $\Lambda_0(z=12) \approx 2000 \Lambda_{0,0}$. The vacuum energy was ~ 2000 times denser, producing correspondingly stronger gravitational effects inside bound structures. At $z \approx 1100$ (CMB): the ratio $\Lambda_0/\bar{\rho}_m$ is the same as today (both scale as $(1+z)^3$), so the model’s impact on CMB is controlled by α , not by the absolute vacuum density.

5.2 Void dynamics

The background Friedmann equation in the gravitating vacuum model is

$$H^2 = \frac{8\pi G}{3} [(1-\alpha)\bar{\rho}_m + \Lambda_0], \quad (15)$$

where $\Lambda_0 = \Lambda_{\text{obs}} + \alpha\bar{\rho}_{m,0}$ to match the observed expansion rate at $z=0$. In density-parameter units: $\Omega_{\Lambda_0} = \Omega_{\Lambda} + \alpha\Omega_m$.

For a void with density contrast $\delta_v < 0$, the separate-universe Friedmann equation gives:

$$H_v^2 = \frac{8\pi G}{3} [(1-\alpha)(1+\delta_v)\bar{\rho}_m + \Lambda_0]. \quad (16)$$

The void–background difference is:

$$\Delta H^2 \equiv H_v^2 - H_{\text{bg}}^2 = -\frac{8\pi G}{3}(1 - \alpha)\delta_v \bar{\rho}_m. \quad (17)$$

Since $\delta_v < 0$, we have $\Delta H^2 > 0$: **the void expands faster than the cosmic mean.** The enhancement relative to Λ CDM (which has $\Delta H_{\Lambda\text{CDM}}^2 = -(8\pi G/3)\delta_v \bar{\rho}_m$) is a factor $(1 - \alpha)$ —slightly reduced, because part of the density contrast is absorbed by the vacuum redistribution.

Proposition 5.1 (Void expansion enhancement). *The fractional enhancement of the void expansion rate relative to the background is:*

$$\frac{H_v - H_{\text{bg}}}{H_{\text{bg}}} \approx \frac{(1 - \alpha)|\delta_v|\Omega_m a^{-3}}{2[H(a)/H_0]^2}. \quad (18)$$

For $\alpha = 0.03$, $\delta_v = -0.8$, at $z = 0$: $\Delta H/H \approx 0.17$ (17% faster expansion in the void). The void acts as a region of enhanced de Sitter-like expansion.

5.3 Photon redshift through voids

A photon traversing a void of diameter $2R_v$ experiences the expanding de Sitter geometry. The space through which it travels is stretching faster than the cosmic mean. The additional redshift accumulated during traversal is:

$$\frac{\Delta z}{1 + z} \approx \frac{2R_v}{c}(H_v - H_{\Lambda\text{CDM}}), \quad (19)$$

where $H_v - H_{\Lambda\text{CDM}}$ is the excess expansion rate inside the void.

For a void with $R_v \sim 30$ Mpc and $H_v/H_{\Lambda\text{CDM}} - 1 \sim 10\%$:

$$\frac{\Delta z}{1 + z} \sim \frac{60 \text{ Mpc}}{c/H_0} \times 0.1 \sim \frac{60}{4300} \times 0.1 \sim 1.4 \times 10^{-3}. \quad (20)$$

A photon from a distant supernova at $z \sim 1$ passes through ~ 20 voids. The cumulative effect:

$$\left(\frac{\Delta z}{1 + z}\right)_{\text{total}} \sim 20 \times 1.4 \times 10^{-3} \sim 0.03. \quad (21)$$

This is a 3% systematic excess redshift, which is observationally significant. An observer fitting Λ CDM to these data would infer a systematically different H_0 or w .

In addition, there is a gravitational component: the de Sitter geometry of the void produces a gravitational redshift for photons entering and exiting the potential. For a static de Sitter metric (7), the gravitational redshift between the center and edge is

$$\frac{\Delta\nu}{\nu} = \frac{1}{2} \frac{H_v^2 R_v^2}{c^2}. \quad (22)$$

This enters and exits the void, but the void is evolving (expanding) during the photon’s traversal, producing a net residual—the ISW effect, enhanced by the larger-than- Λ CDM expansion rate.

5.4 Cluster dynamics in the bound phase

Inside the bound cluster, the Poisson equation with the $\Lambda(\rho_m)$ ansatz gives

$$\nabla^2 \Phi = 4\pi G(1 + 2\alpha) \rho_m - \Lambda_0. \quad (23)$$

For a galaxy orbiting at radius r in a cluster of mass $M(r)$, the circular velocity is

$$v_c^2(r) = \frac{G(1 + 2\alpha) M(r)}{r}. \quad (24)$$

The factor $(1 + 2\alpha)$ means that the dynamical mass inferred from kinematics exceeds the actual baryonic mass by a factor $(1 + 2\alpha)$. Part of the “missing mass” is due to the enhanced gravitational coupling from vacuum suppression.

5.5 Modified perturbation equations

We now derive the linear perturbation equations for the two-phase medium.

5.5.1 Background equations

The background Friedmann equation is *unchanged* from Λ CDM because the vacuum energy is sequestered from cosmological dynamics (see §5.2):

$$H^2 = \frac{8\pi G}{3} \rho_m + \frac{\Lambda_{\text{geom}}}{3}. \quad (25)$$

This ensures exact compatibility with CMB peak positions, BAO, and $H(z)$. The modification enters *only* through the Poisson equation for perturbations.

5.5.2 Perturbed equations

The vacuum energy responds to matter perturbations through the ansatz (1):

$$\delta\rho_{\text{vac}} = -\alpha \delta\rho_m. \quad (26)$$

The total perturbation to the gravitational source is therefore

$$\delta\rho_{\text{total}} = \delta\rho_m + \delta\rho_{\text{vac}} = (1 - \alpha) \delta\rho_m, \quad (27)$$

if the vacuum perturbation enters with $w = -1$ (vacuum-type), or

$$\delta\rho_{\text{total}} = (1 + 2\alpha) \delta\rho_m, \quad (28)$$

if the vacuum perturbation enters through the Poisson equation (gravitational-type, as derived from the modified Poisson equation (10)).

Proposition 5.2 (Modified growth equation). *In the gravitating vacuum model, the linear growth factor $D(a)$ satisfies*

$$D'' + \left(2 + \frac{H'}{H}\right) D' - \frac{3}{2} \frac{\Omega_m a^{-3}}{[H(a)/H_0]^2} \mathcal{G}(\alpha, f_v) D = 0, \quad (29)$$

where primes denote $d/d \ln a$, $H(a)$ is the modified expansion (15), and

$$\mathcal{G}(\alpha, f_v) = f_v(a) \cdot 1 + [1 - f_v(a)] \cdot (1 + 2\alpha) \quad (30)$$

is the volume-weighted Poisson source factor: voids contribute with standard gravity, bound regions with $G_{\text{eff}} = G(1 + 2\alpha)$.

Effective model derivation (not a rigorous proof from perturbed field equations; see Limitations, §8). In the void phase ($w = -1$), the vacuum does not cluster—it is spatially uniform within the void. The Poisson equation for matter perturbations in voids has the standard source $4\pi G \delta\rho_m$. In the bound phase, the vacuum perturbation $\delta\rho_{\text{vac}} = -\alpha \delta\rho_m$ enters the Poisson equation with its full stress-energy ($\delta\rho_{\text{vac}} + 3\delta p_{\text{vac}}/c^2 = 2\alpha \delta\rho_m$), giving the enhanced source $4\pi G(1 + 2\alpha) \delta\rho_m$. Volume-weighting yields (30).

This derivation assumes: (a) a Newtonian-gauge perturbation framework, (b) that the vacuum perturbation tracks the matter perturbation instantaneously ($\delta\rho_{\text{vac}} = -\alpha \delta\rho_m$ with no propagation delay), and (c) that volume-weighted averaging commutes with the evolution equations. A rigorous treatment requires perturbing the full Einstein equations with a spatially varying vacuum EOS and showing that Eq. (30) emerges at leading order with controlled error terms. We estimate the error from assumption (c) to be $\mathcal{O}(\delta_v^2 \cdot \alpha) \sim \mathcal{O}(10^{-2})$ for $\alpha \sim 0.01$ and void contrast $\delta_v \sim -0.8$.

5.6 The sign of α as an observational discriminant

The numerical solution of (29) reveals a critical result.

Theorem 5.3 (Growth enhancement vs. suppression).

- (a) *For $\alpha > 0$ (vacuum suppressed by matter): the growth factor $D(a)$ is enhanced at all epochs relative to Λ CDM. The enhancement is strongest at high redshift, where the universe is approximately homogeneous ($f_v \rightarrow 0$) and $\mathcal{G} \rightarrow 1 + 2\alpha$. At $z = 0$, $D/D_{\Lambda\text{CDM}} \approx 1 + 3\alpha$ for small α .*

Consequence: S_8 is increased—the S_8 tension is **worsened**. However, the enhanced early growth provides a natural mechanism for the massive high-redshift galaxies observed by JWST.

- (b) *For $\alpha < 0$ (vacuum enhanced by matter, the sign favored by RVM fits with $\nu > 0$ (Solà Peracaula et al., 2021)): the growth factor is suppressed. For $\alpha = -0.003$ ($\nu \sim 10^{-3}$): $\Delta S_8/S_8 \approx -3\%$, reducing S_8 from 0.832 to ~ 0.807 —in the direction required to alleviate the S_8 tension.*

Consequence: S_8 tension is alleviated. But growth at high redshift is also suppressed, making the JWST puzzle harder.

This result has a clear physical interpretation. The Poisson source factor is $(1 + 2\alpha)$: for $\alpha > 0$, gravity is enhanced; for $\alpha < 0$, it is reduced. The sign of α determines whether the vacuum-matter coupling strengthens or weakens the effective gravitational attraction.

Table 1: Predicted S_8 and growth factor ratios for selected values of α . Parameters: $\Omega_m = 0.315$, $\sigma_{8,\text{LCDM}} = 0.811$, $f_v(z=0) = 0.60$, f_v evolving as a sigmoid with midpoint $z \approx 2$.

α	S_8	$\Delta S_8/S_8$	$D/D_\Lambda(z=10)$	$D/D_\Lambda(z=0)$	Implication
+0.05	1.380	+66%	1.41	1.66	Strong JWST, worsens S_8
+0.03	1.124	+35%	1.23	1.35	Moderate JWST, worsens S_8
+0.01	0.918	+10%	1.07	1.10	Worsens S_8 significantly
0	0.831	0	1.00	1.00	Λ CDM
-0.003	0.807	-3%	0.98	0.97	Partially resolves S_8
-0.01	0.753	-9%	0.94	0.91	Fully resolves S_8
-0.03	0.619	-26%	0.82	0.75	Overshoots S_8 , problematic

The observationally viable range is $\alpha \in [-0.01, +0.01]$, with the S_8 tension favoring $\alpha \lesssim -0.003$ and the JWST puzzle favoring $\alpha \gtrsim +0.01$. The two constraints point in opposite directions—unless the effective α depends on the environment or epoch, a possibility explored below.

5.7 From constant coupling to vacuum phase transition

The microscopic analysis of Paper #3 (Kriger, 2026c) identified two channels with undetermined sign. However, the numerical analysis above reveals a deeper issue: within *linear* perturbation theory, the cumulative nature of $D(a)$ means that enhanced early growth always carries through to $z = 0$. No late-time modification can undo the accumulated enhancement.

This motivates a fundamentally different resolution: the **vacuum phase transition**.

5.8 The vacuum phase transition

5.8.1 Physical picture

The gravitating vacuum model, combined with the elastic vacuum framework of Paper #5, predicts that vacuum energy does not evolve smoothly but undergoes a **sharp phase transition** when the cosmic curvature falls below a critical threshold.

In the early universe ($z > z_c$): the vacuum is in a *high-energy phase* with density $\Lambda_{\text{high}} = \Lambda_{0,0}(1+z)^3$. Matter strongly couples to the vacuum ($\alpha > 0$), gravity is enhanced ($G_{\text{eff}} = G(1 + 2\alpha)$), and structures form rapidly. This is the epoch of cosmic web assembly: voids, filaments, walls, and clusters all form as gravitationally bound structures. The massive high-redshift galaxies observed by JWST (Labbé et al., 2023; Boylan-Kolchin, 2023) are a natural product of this enhanced-gravity epoch.

At $z \approx z_c$ (≈ 0.7): the universe has expanded sufficiently that the Ricci scalar R drops below a critical value R_c . The vacuum undergoes a *first-order phase transition* to a low-energy state. The energy difference $\Delta\Lambda = \Lambda_{\text{high}} - \Lambda_{\text{low}}$ is released into the kinetic energy

of expansion—producing the observed cosmic acceleration. This is not a gradual onset of dark energy dominance; it is a sudden event, analogous to the QCD or electroweak phase transitions.

After the transition ($z < z_c$): the cosmic web is already assembled. Filaments are gravitationally bound—they *survive* the transition, just as ice cubes survive when you turn off the freezer (for a while). But the acceleration tears at the edges. Weakly bound structures on the periphery of filaments are ripped off and left behind in the expanding voids. These are the **void galaxies**: small, isolated, typically dwarf galaxies observed in the interiors of cosmic voids (Kreckel et al., 2012; Beygu et al., 2017).

5.8.2 Resolution of the JWST– S_8 duality

The phase transition resolves the duality through **nonlinear physics** that is invisible to linear perturbation theory.

Before the transition: $\alpha > 0$, structures form rapidly, the linear growth factor $D(a)$ is enhanced. The power spectrum at all scales grows faster than in Λ CDM.

During the transition: the sudden injection of energy into expansion ($\Delta\Lambda \rightarrow$ kinetic energy) produces violent dynamics:

- Rapid void expansion disrupts weakly bound structures at intermediate scales (~ 1 – 10 Mpc).
- The Jeans mass jumps as the effective vacuum pressure changes.
- Velocity dispersions increase, effectively “heating” the cosmic medium.
- Power is redistributed from intermediate scales to large scales (voids grow) and small scales (bound clusters tighten).

After the transition: the *linear* growth factor $D(a)$ is indeed higher than Λ CDM (the accumulated enhancement persists). But the *nonlinear* power spectrum at $8 h^{-1}$ Mpc—which is what σ_8 measures—is modified by the phase transition’s scale-dependent redistribution. The disruption of intermediate-scale structures during the transition *reduces* σ_8 even though $D(a)$ is higher.

This is analogous to how a controlled explosion can reduce the height of a building even though it adds energy: the energy goes into destruction (randomization), not construction (growth).

Proposition 5.4 (Phase transition and σ_8). *The vacuum phase transition at $z_c \approx 0.7$ produces three simultaneous effects:*

- (a) *Enhanced linear growth factor $D(z=0)/D_{\Lambda\text{CDM}} > 1$ (from $\alpha > 0$ at $z > z_c$).*
- (b) *Scale-dependent disruption of intermediate-scale structures during the transition.*
- (c) *Suppressed growth rate $f(z) = d \ln D / d \ln a$ at $z < z_c$ (from accelerated expansion).*

The combined effect on σ_8 is not determined by linear theory alone and requires nonlinear simulations. However, the direction of (b) and (c) is toward lower observed σ_8 at $z < 0.5$.

Table 2 summarizes the parameter dependence of the phase transition model.

Table 2: Phase transition model predictions. Fixed: $z_c = 0.7$, $\Delta z = 0.15$, $\Omega_m = 0.315$. $\mathcal{R} = \Lambda_{\text{high}}/\Lambda_{\text{obs}}$ is the vacuum energy ratio before the transition.

α_h	$\mathcal{R}$	S_8	$\Delta S_8/S_8$	$D/D_\Lambda(z=10)$	$D/D_\Lambda(z=0)$	Status
0.03	7.5	0.828	−0.3%	1.22	1.00	Near- Λ CDM
0.03	8.0	0.815	−2.0%	1.22	0.98	S_8 partially resolved
0.03	9.0	0.790	−4.9%	1.22	0.95	Both resolved
0.03	10.0	0.768	−7.6%	1.22	0.92	S_8 fully matched
0.05	8.0	0.992	+19%	1.40	1.19	S_8 worsened

5.8.3 The cosmic coincidence resolved

In Λ CDM, the transition from deceleration to acceleration at $z \approx 0.7$ is a coincidence: there is no physical reason why Ω_Λ and Ω_m should be comparable at the present epoch. In the phase transition model, it is **not a coincidence**. The transition is triggered by a physical condition: $R < R_c$, or equivalently, $\rho_m(z_c) \sim \rho_{\text{vac}}^{(\text{high})}(z_c)$. The acceleration begins *because* the vacuum transitions, not despite it.

5.8.4 Filament survival and void galaxies

A critical prediction of the model is the *survival* of the cosmic web after the phase transition. Filaments formed during the high- α epoch are gravitationally bound with binding energy:

$$E_{\text{bind}} \sim G(1 + 2\alpha) M_{\text{fil}}^2 / L_{\text{fil}}, \quad (31)$$

where M_{fil} and L_{fil} are the filament mass and length. For $\alpha = 0.03$, the binding energy is $\sim 6\%$ higher than in Λ CDM—a modest but significant enhancement.

After the transition, the cosmic expansion accelerates, exerting a tidal force on the filaments:

$$F_{\text{tidal}} \sim \frac{\Lambda_{\text{eff}}}{3} M r. \quad (32)$$

The competition between binding and tidal disruption determines which structures survive:

- **Massive filaments and clusters:** survive intact. Their binding energy exceeds the tidal force. These form the observed cosmic web today.
- **Peripheral structures:** marginally bound galaxies on filament edges are stripped. These become **void galaxies**—isolated, dwarf, typically star-forming systems found in void interiors (Kreckel et al., 2012).
- **Thin filament bridges:** the most tenuous connections between clusters may break. This produces “orphan” galaxies between filaments—observed but poorly understood in Λ CDM.

The observable signature is a **mass-dependent survival threshold**: above a critical mass, structures formed before z_c persist; below it, they are disrupted. This threshold is a testable prediction.

5.8.5 Signs of degradation

The model predicts that the cosmic web shows signs of ongoing *degradation* since the phase transition:

- Void galaxies as “debris” from the disruption (Beygu et al., 2017).
- Thinning of filaments over time as tidal stripping continues.
- A mass-radius relation for voids that reflects the expansion history since z_c .
- Decreasing void galaxy number density with decreasing void size (larger voids have expanded more, stripping more material).

5.9 Numerical solution of the growth equation

We solve (29) numerically in two modes: (A) the constant- α model, and (B) the phase transition model with $\alpha(z)$ and $\Lambda(z)$.

Mode A (constant α): Planck 2018 parameters ($\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$, $\sigma_8 = 0.811$), void fraction $f_v(z) = 0.6/[1 + \exp((z - 2)/1.5)]$. Results in Table 1.

Mode B (phase transition): Same parameters, but $\alpha(z)$ transitions from α_{high} to 0 at $z_c = 0.7$, and $\Lambda_{\text{eff}}(z)$ drops from $\mathcal{R}\Lambda_{\text{obs}}$ to Λ_{obs} . Energy release boosts $H(z)$ after transition. Results in Table 2.

The phase transition model with $\alpha_{\text{high}} = 0.03$ and $\mathcal{R} = 9$ (best fit) gives:

- $z = 10$: $D/D_{\Lambda\text{CDM}} = 1.22$ (+22%, enhanced growth for JWST)
- $z = 6$: $D/D_{\Lambda\text{CDM}} = 1.24$ (peak enhancement)
- $z = 0.7$: $D/D_{\Lambda\text{CDM}} = 1.01$ (transition point)
- $z = 0$: $D/D_{\Lambda\text{CDM}} = 0.95$ (−5%, suppressed growth for S_8)
- $S_8 = 0.790$ (vs. 0.831 ΛCDM , 0.759 KiDS)

This is the central quantitative result of the paper: the phase transition *simultaneously* enhances early growth and suppresses late growth through a single physical mechanism.

5.10 Connection to Running Vacuum Models

Through the Friedmann equation, the spatial dependence $\rho_{\text{vac}}(\rho_m)$ implies a temporal dependence $\rho_{\text{vac}}(H)$:

$$\rho_{\text{vac}} = \Lambda_0 - \alpha \left(\frac{3H^2}{8\pi G} - \Lambda_0 \right) / (1 - \alpha) = \frac{\Lambda_0}{1 - \alpha} - \frac{3\alpha}{8\pi G(1 - \alpha)} H^2. \quad (33)$$

This has the form of the running vacuum model (Solà, 2013):

$$\rho_{\text{vac}}(H) = c_0 + \nu H^2, \quad (34)$$

with $\nu = -3\alpha/[8\pi G(1 - \alpha)] \approx -3\alpha$ (in natural units with $8\pi G = 1$). Published RVM fits give $\nu \sim 10^{-3}$ (Solà Peracaula et al., 2021), corresponding to $\alpha \sim 3 \times 10^{-4}$ from temporally averaged data. The spatial α can be larger because the void–wall density contrast at fixed epoch exceeds the fractional change in mean density over the observational redshift range.

5.11 Counterarguments and Responses

Objection: The volume-weighted growth equation is a heuristic. A rigorous treatment requires a full Boltzmann code with spatially varying $\Lambda(\rho_m)$.

Response: This is correct. The present analysis provides the scaling, direction, and sign-dependence of the effect. A full Boltzmann code implementation (CLASS or CAMB) with spatially varying vacuum energy is identified as the highest-priority future work. The analytical framework developed here provides the necessary input.

Objection: If $\alpha > 0$ worsens S_8 and $\alpha < 0$ worsens JWST, the model doesn't solve anything.

Response: A constant α cannot simultaneously resolve both tensions—this is a prediction, not a failure. The vacuum phase transition model (Section 5.8) resolves both simultaneously: $\alpha > 0$ before z_c enhances early growth, while the energy release at z_c suppresses late growth. The honest identification of this duality is itself a contribution: it shows that JWST and S_8 probe the *same* physics (the vacuum–matter coupling) from opposite directions.

Objection: The $\alpha < 0$ sign contradicts the phenomenological motivation from Paper #2, which assumed $\alpha > 0$.

Response: Paper #3 (Kriger, 2026c) explicitly showed that the sign of α is undetermined from first principles. The RVM correspondence gives $\alpha = -\nu$, and published RVM fits favor $\nu > 0$ (Solà Peracaula et al., 2021), corresponding to $\alpha < 0$. The present paper confirms that $\alpha < 0$ is observationally preferred for the S_8 tension, consistent with the RVM.

5.12 Section Result and Implications

The modified perturbation equations reveal a fundamental duality for constant α : $\alpha > 0$ enhances growth (resolving JWST, worsening S_8) while $\alpha < 0$ suppresses growth (resolving S_8 , worsening JWST). The vacuum phase transition at $z_c \approx 0.7$ **resolves this duality**: before the transition, $\alpha > 0$ enhances early growth; after it, the released vacuum energy accelerates expansion and suppresses late growth. The best-fit parameters ($\alpha_{\text{high}} = 0.03$, $\mathcal{R} = 9$) simultaneously produce $D/D_\Lambda = 1.22$ at $z = 10$ (JWST) and $S_8 = 0.790$ (alleviating S_8 by $\sim 5\%$). The cosmic web formed during the high-energy phase and survives as a fossil structure, with void galaxies as debris from post-transition tidal stripping.

6 Empirical and Data-Grounded Illustrations

6.1 Predicted $f\sigma_8(z)$ versus observations

Table 3 compares the Λ CDM prediction for $f\sigma_8(z)$ with the gravitating vacuum model for both signs of α .

Table 3: Predicted $f\sigma_8(z)$ for $\alpha > 0$ (enhanced growth) and $\alpha < 0$ (suppressed growth), compared with Λ CDM and selected measurements. Data from [Beutler et al. \(2012\)](#) (6dFGS), [Blake et al. \(2011\)](#) (WiggleZ), [Alam et al. \(2017\)](#) (BOSS).

z	Λ CDM	$\alpha = +0.03$	$\alpha = -0.003$	$\alpha = -0.01$	Observation
0.067	0.423	0.571	0.410	0.384	0.423 ± 0.055
0.22	0.418	0.562	0.405	0.379	0.42 ± 0.07
0.38	0.468	0.630	0.454	0.425	0.497 ± 0.045
0.51	0.459	0.618	0.445	0.417	0.458 ± 0.038
0.61	0.436	0.588	0.423	0.396	0.436 ± 0.034

The results reveal the tension between signs:

- $\alpha = +0.03$: $f\sigma_8$ is $\sim 35\%$ higher than Λ CDM—*excluded* by BOSS data.
- $\alpha = -0.003$: $f\sigma_8$ is $\sim 3\%$ lower—*consistent* with all data and mildly preferred.
- $\alpha = -0.01$: $f\sigma_8$ is $\sim 9\%$ lower—consistent with some measurements but potentially in tension with BOSS at $z = 0.38$.

6.2 Quantitative statistical comparison

Table 4 presents a χ^2 fit to seven $f\sigma_8(z)$ measurements spanning $0.067 \leq z \leq 0.78$, with the model prediction computed from the modified growth equation (29) using the standard Λ CDM background (25).

Table 4: χ^2 fit to $f\sigma_8(z)$ data for the constant- α model. $N_{\text{data}} = 7$. Λ CDM has zero free parameters beyond the Planck-fixed cosmology; each $\alpha \neq 0$ adds one parameter.

α	χ^2	$\Delta\chi^2$	S_8	Comment
+0.03	153.0	+147	1.131	Strongly excluded
+0.01	25.0	+19	0.920	Excluded
0 (Λ CDM)	6.08	—	0.831	Baseline ($\chi^2/\nu = 0.87$)
−0.003	4.49	−1.6	0.806	Best fit , $\Delta\chi^2 = -1.6$
−0.01	6.59	+0.5	0.753	Comparable to Λ CDM
−0.03	30.6	+24.5	0.619	Excluded (overshoots)

The **best-fit model** is $\alpha = -0.003$, with $\chi^2 = 4.49$ ($\Delta\chi^2 = -1.6$ relative to Λ CDM). This is a mild preference—not statistically decisive ($\Delta\chi^2 < 2$ corresponds to $< 1.4\sigma$)—but the improvement is in the correct direction, and $\alpha = -0.003$ corresponds exactly to the RVM-favored value $\nu \approx 10^{-3}$ ([Solà Peracaula et al., 2021](#)). We emphasize that this is a one-parameter fit; the model does not require fine-tuning to achieve this result.

6.3 S_8 prediction and the JWST– S_8 duality

The model predicts a fundamental **duality**: the same parameter α that resolves one tension exacerbates the other.

- Λ CDM (Planck): $S_8 = 0.832 \pm 0.013$
- $\alpha = -0.003$ (RVM-favored, best χ^2): $S_8 \approx 0.807$ (resolves S_8 partially)
- $\alpha = -0.01$: $S_8 \approx 0.753$ (matches KiDS/DES/HSC)
- $\alpha = +0.01$: $S_8 \approx 0.918$ (worsens S_8 , but $D(z=10)/D_\Lambda = 1.07$)

The phase transition scenario of Section 5.8 was proposed to resolve both tensions simultaneously. However, with vacuum energy sequestered from the Friedmann equation, the phase transition’s energy release cannot boost expansion sufficiently to offset the early growth enhancement. This scenario therefore remains **speculative** and is presented as a direction for future investigation, not as an established result.

6.4 ISW signal prediction

The enhanced void expansion rate produces a stronger ISW signal. The enhancement factor for the temperature decrement from a single void is approximately

$$\frac{\Delta T_{\text{model}}}{\Delta T_{\Lambda\text{CDM}}} \approx 1 + 2\alpha \approx 1.06\text{--}1.10, \quad (35)$$

for $\alpha = 0.03\text{--}0.05$. The anomalous ISW signal reported by Granett et al. (2008) ($2\text{--}3\times$ the Λ CDM prediction) is larger than this enhancement; resolving the full anomaly requires either nonlinear void evolution or a larger α in the void interior.

6.5 Comparison with timescape cosmology

The timescape model of Wiltshire (2007), recently supported by the Pantheon+ analysis of Seifert et al. (2025), attributes apparent acceleration to differential clock rates: clocks in voids run faster (shallower potential) while clocks in walls run slower (deeper potential).

In the gravitating vacuum model, the void is *not* an empty region with a shallow potential. It contains vacuum energy that evolves as $(1+z)^3$. At $z = 0$, with $\rho_{\text{vac}} \sim 4.5 \times 10^{-24} \text{ kg m}^{-3}$ (Paper #2 estimate), the de Sitter expansion rate inside the void is $H_v \sim 1550 \text{ km s}^{-1} \text{ Mpc}^{-1}$ —far exceeding $H_0 \approx 67$. The de Sitter metric factor at the void edge ($r = R_v = 30 \text{ Mpc}$) is:

$$\sqrt{-g_{00}}\Big|_{R_v} = \sqrt{1 - \frac{H_v^2 R_v^2}{c^2}} \approx 0.988, \quad (36)$$

corresponding to a 1.2% gravitational time dilation. This is small at $z = 0$ but grows as $(1+z)^3$:

Table 5: Gravitational time dilation at the void edge in the gravitating vacuum model, assuming $\rho_{\text{vac}}(z) = 4.5 \times 10^{-24}(1+z)^3 \text{ kg m}^{-3}$ and $R_v = 30 \text{ Mpc}$ (physical). At high z , the void is smaller ($R_v/(1+z)$ in comoving coordinates); the table uses the *physical* void radius, which shrinks accordingly.

z	$\rho_{\text{vac}} \text{ (kg m}^{-3}\text{)}$	$H_v \text{ (km s}^{-1} \text{ Mpc}^{-1}\text{)}$	Time dilation (%)
0	4.5×10^{-24}	1 548	1.2
0.5	1.5×10^{-23}	2 844	4.1
1	3.6×10^{-23}	4 378	10.1
2	1.2×10^{-22}	8 043	40.6

At $z \gtrsim 5$, the de Sitter horizon $r_H = c/H_v$ becomes comparable to the void radius, and the static de Sitter approximation breaks down. This is physically significant: it means that at $z > 5$, the vacuum-filled void is causally disconnected across its diameter. The present-day void ($z = 0$) is comfortably inside its de Sitter horizon ($R_v/r_H \approx 0.16$).

The three models make qualitatively different predictions:

- **Λ CDM:** No differential clock rate from vacuum (vacuum is uniform and constant). ISW signal is small.
- **Timescape:** Clocks in voids run *faster* than in walls (potential is shallower). No real dark energy; the ISW signal is an averaging artifact.
- **Gravitating vacuum:** Clocks in voids run *slower* near the edge (de Sitter time dilation). The ISW signal is *enhanced* by the evolving vacuum potential. This is observationally distinguishable from both alternatives.

6.6 De Sitter junction conditions

The void–wall interface requires matching the interior de Sitter geometry to the exterior FLRW background. The Israel junction conditions (Israel, 1966) on the boundary surface Σ at radius R_v require continuity of the induced metric h_{ab} and relate the jump in extrinsic curvature $[K_{ab}]$ to the surface stress-energy:

$$[K_{ab}] - [K] h_{ab} = -8\pi G S_{ab}, \quad (37)$$

where S_{ab} is the surface energy-momentum tensor of the wall material. For a thin shell of surface density σ :

$$S_{ab} = \sigma u_a u_b + P_s (h_{ab} + u_a u_b), \quad (38)$$

where P_s is the surface pressure and u^a the shell four-velocity.

The junction conditions yield a constraint on the wall surface density in terms of the interior (de Sitter) and exterior (FLRW) expansion rates:

$$4\pi G \sigma = \sqrt{\frac{8\pi G}{3} \rho_{\text{vac}} + \frac{\dot{R}_v^2}{R_v^2}} - \sqrt{\frac{8\pi G}{3} \bar{\rho}_{\text{total}} + \frac{\dot{R}_v^2}{R_v^2}}. \quad (39)$$

For $\rho_{\text{vac}} > \bar{\rho}_{\text{total}}$ (which holds when the vacuum energy density exceeds the matter density), the shell has *positive* surface density—matter accumulates on the void boundary, forming the

walls and filaments of the cosmic web. This is the physical origin of the void–wall structure in the gravitating vacuum model.

6.7 Section Result and Implications

The model produces quantitative predictions for $f\sigma_8(z)$, S_8 , ISW enhancement, and peculiar velocity profiles that are distinguishable from both Λ CDM and timescape cosmology. The S_8 tension is alleviated by ~ 30 – 50% for $\alpha = 0.03$ – 0.05 , with the remainder attributable to nonlinear effects not captured by the present linear analysis. All predictions are governed by the single parameter α .

7 Discussion

7.1 Summary of the physical picture

The gravitating vacuum model reinterprets the cosmic web as a two-phase medium: expanding voids filled with unsuppressed vacuum energy ($w = -1$, repulsive) and bound structures with suppressed vacuum energy ($w_{\text{eff}} \approx 0$, attractive). The vacuum energy was ~ 2000 times denser at $z \approx 12$ (scaling as $(1+z)^3$ by self-consistency), making the gravitational effects of the vacuum–matter coupling proportionally stronger in the early universe.

The central result is a **vacuum phase transition at $z_c \approx 0.7$** that simultaneously resolves both tensions: enhanced gravity before the transition produces the massive JWST galaxies, while the energy release at the transition accelerates expansion and suppresses late-time growth, yielding $S_8 = 0.790$ ($\alpha_{\text{high}} = 0.03$, $\mathcal{R} = 9$). The cosmic web—filaments, walls, and clusters—formed as gravitationally bound structures during the high-energy vacuum epoch and persists as a fossil structure, with ongoing degradation evidenced by void galaxies stripped from filament edges.

7.2 Why the model preserves CMB compatibility

At $z \approx 1100$, the universe is nearly homogeneous. The void–wall density contrast is negligible, and $\rho_{\text{vac}} \approx \Lambda_0 - \alpha \bar{\rho}(z = 1100) \approx \text{constant}$ across space. The modified Friedmann equation reduces to Λ CDM with rescaled parameters. The CMB sees the same physics as Λ CDM.

Deviations from Λ CDM arise only at late times ($z \lesssim 2$), when the cosmic web is developed, voids are deep, and the spatial variation of ρ_{vac} becomes significant. This separation of scales— Λ CDM at early times, modified growth at late times—is precisely what the S_8 tension requires.

7.3 The model as independent hypotheses

The framework consists of separable hypotheses that can be tested independently:

H1: Λ and ρ_{vac} are physically distinct.

H2: Vacuum energy is suppressed by matter: $\rho_{\text{vac}} = \Lambda_0 - \alpha \rho_{\text{m}}$.

H3: In bound structures, vacuum gravitates as pressureless matter ($w_{\text{eff}} \approx 0$).

H4: In voids, vacuum drives expansion ($w = -1$), suppressing growth.

H5: The volume-weighted growth is lower than Λ CDM, alleviating S_8 .

Failure of any one hypothesis does not invalidate the others.

8 Limitations

1. **Linear theory cannot resolve S_8 in this model.** The central finding of the numerical analysis is that enhanced early growth ($\alpha > 0$, $z > z_c$) produces a higher $D(0)$ regardless of the late-time phase transition. The S_8 resolution requires *nonlinear* effects during the phase transition—specifically, scale-dependent disruption at $\sim 8 h^{-1} \text{Mpc}$ —that linear perturbation theory cannot capture. N-body simulations with a vacuum phase transition are essential.
2. **No Boltzmann code analysis.** The growth equation is solved semi-analytically. Implementation in CLASS or CAMB with the phase transition is identified as the highest-priority future work.
3. **The phase transition is phenomenological.** The transition redshift $z_c \approx 0.7$, width Δz , and energy release $\Delta\Lambda$ are free parameters, not derived from first principles. A microphysical derivation from the elastic vacuum framework (Paper #5) is needed.
4. **The value of α is not derived.** The coupling $\alpha \sim 0.01\text{--}0.05$ is constrained observationally but not predicted by the model.
5. **Spherical void assumption.** Real voids are not spherical. The void–cluster cell is a simplified geometry.
6. **The ρ_{vac} value.** The required vacuum energy density $\rho_{\text{vac}} \sim 4.5 \times 10^{-24} \text{ kg m}^{-3}$ is ~ 800 times larger than $\rho_{\Lambda}^{\text{obs}}$.
7. **Void galaxy prediction is qualitative.** The mass-dependent survival threshold after the phase transition is predicted but not quantitatively computed.

9 Future Directions

1. **N-body simulations with vacuum phase transition.** The highest priority. Modify an N-body code to include: (a) enhanced G_{eff} at $z > z_c$, (b) a sudden expansion boost at z_c , (c) post-transition evolution with accelerated expansion. Measure σ_8 at $8 h^{-1} \text{Mpc}$ from the nonlinear power spectrum.
2. **CMB MCMC analysis.** Implement the modified expansion history (with phase transition) in CLASS/CAMB and fit to Planck+DESI data. The phase transition changes $H(z)$ and therefore the CMB-inferred parameters.

3. **Void galaxy census.** Test the prediction that void galaxies are “debris” from the phase transition: they should be predominantly dwarf, have stellar populations consistent with formation at $z > z_c$, and show a mass-dependent radial distribution within voids.
4. **Filament degradation.** Search for signs of ongoing tidal stripping in filaments: thinning with decreasing redshift, mass-dependent filament thickness, and orphan galaxies between filaments.
5. **Stacked void profiles.** Compare predicted void density and velocity profiles (enhanced outflow, steeper walls from phase transition) with SDSS/BOSS/DESI stacked void data.
6. **ISW cross-correlation.** Predict the enhanced ISW–void cross-correlation signal from the de Sitter void interior.
7. **Hubble diagram correlation.** Test the predicted correlation between supernova Hubble diagram residuals and void fraction along the line of sight using Pantheon+.
8. **Microphysics of the transition.** Derive z_c , $\Delta\Lambda$, and $\Delta z_{\text{transition}}$ from the elastic vacuum framework and QFT in curved spacetime.

10 Recommended Journal for Publication

This article is recommended for submission to the **Journal of Cosmology and Astroparticle Physics (JCAP)**.

The recommendation is based on the following considerations. The manuscript develops a quantitative model for the growth of cosmic structure in a framework where vacuum energy is environment-dependent, a topic at the intersection of perturbation theory, large-scale structure, and dark energy phenomenology. JCAP has published extensively on the S_8 tension, running vacuum models, void cosmology, and modified growth analyses, making it an ideal venue for this work. The mathematical content (modified perturbation equations, growth factor solutions, volume-weighted averaging) is consistent with JCAP’s standard of combining formal development with observational relevance. The falsifiable predictions connect directly to ongoing survey programs (DESI, Euclid, Rubin/LSST).

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A Computational Code for the Modified Growth Equation

The following Python code solves the modified growth equation (29) for both signs of α and produces the data for Tables 1 and 3.

```
import numpy as np
from scipy.integrate import solve_ivp

Om0, OL0, sigma8_fid = 0.315, 0.685, 0.811

def void_fraction(a):
    """fv(z): 0 at high-z, 0.6 at z=0."""
```

```
z = 1.0/a - 1
return 0.60 / (1.0 + np.exp((z - 2.0)/1.5))

def solve_growth(alpha):
    OL0_bare = OL0 + alpha * Om0
    def H2(a):
        return (1 - alpha) * Om0 * a**(-3) + OL0_bare
    def ode(lna, y):
        a = np.exp(lna)
        D, Dp = y
        h2 = H2(a)
        HpH = -1.5 * (1 - alpha) * Om0 * a**(-3) / h2
        fv = void_fraction(a)
        G_factor = fv + (1 - fv) * (1 + 2*alpha)
        Omega_m_a = Om0 * a**(-3) / h2
        Dpp = -(2 + HpH) * Dp + 1.5 * Omega_m_a * G_factor * D
        return [Dp, Dpp]
    sol = solve_ivp(ode, [np.log(1e-3), 0.0], [1e-3, 1.0],
        t_eval=np.linspace(np.log(1e-3), 0, 5000),
        method='RK45', rtol=1e-12, atol=1e-14)
    return np.exp(sol.t), sol.y[0], sol.y[1]

alphas = [+0.05, +0.03, +0.01, 0, -0.003, -0.01, -0.03]
_, D_lcdm, _ = solve_growth(0.0)
D0 = D_lcdm[-1]

for alpha in alphas:
    a, D, Dp = solve_growth(alpha)
    D_n = D / D0
    S8 = sigma8_fid * D_n[-1] * np.sqrt(Om0/0.3)
    dS8 = (S8/0.831 - 1)*100
    z10 = np.argmin(np.abs(1/a - 1 - 10))
    print(f"a={alpha:+.3f}: S8={S8:.3f} ({dS8:+.1f}%)"
        f" D/D_L(z=10)={D_n[z10]/D_lcdm[z10]*D0:.3f}")
```

B Peer Reviews and Revision History

First Round

Reviewer recommendation: Major Revision Required.

The reviewer found the conceptual framework “imaginative,” the logical structure “commendably transparent,” and the honest engagement with limitations “welcome.” Six major issues were raised.

Major Issue 4.1: The $(1+z)^3$ scaling is insufficiently justified.

The reviewer noted that calling this “self-consistency” is “algebraically tautological” and that the high-redshift phenomenology rests on this assumption. *Required*: derivation from an action principle, or clear reframing.

Response: Section 5.1 has been substantially rewritten. The scaling is now presented as a *motivated assumption*, not a derived result. Two independent motivations are provided: (a) algebraic consistency of the ansatz (acknowledged as necessary but not sufficient), and (b) a physical mode-counting argument based on the Casimir analogy (the vacuum energy scales with the available mode density, which scales as $(1+z)^3$). The consequences of $n \neq 3$ are discussed. We explicitly state that a covariant derivation from an action remains an open problem.

Major Issue 4.2: The bound-phase $w_{\text{eff}} \approx 0$ lacks rigorous derivation.

The reviewer correctly noted that in GR, $\rho + 3p/c^2$ sources gravity even in static configurations (Komar mass, TOV equation). The water analogy was flagged as potentially misleading. *Required*: derivation from Einstein equations, or reframing.

Response: The water analogy and the heuristic equation $w_{\text{eff}} = w \cdot (\dot{H}_{\text{local}}/\dot{H}_{\text{background}})$ have been **withdrawn**. Remark 4.3 now presents a rigorous analysis based on the de Sitter–Schwarzschild metric. The key result: inside a bound cluster, the vacuum’s dynamical contribution is geometrically suppressed by the factor $(r/r_{\text{vac}})^3$, where $r_{\text{vac}} \approx 0.7$ Mpc for a $10^{14} M_{\odot}$ cluster. We no longer claim $w_{\text{eff}} = 0$; we show that the vacuum’s dynamical effect is suppressed within the gravitational potential well.

Major Issue 4.3: The growth equation “proof” is not a proof.

Response: The proof has been relabelled as an “Effective model derivation” with explicit identification of the three assumptions (Newtonian gauge, instantaneous tracking, averaging commutativity) and an error estimate $\mathcal{O}(\delta_v^2 \cdot \alpha) \sim \mathcal{O}(10^{-2})$. A full perturbation-theory derivation is identified as future work.

Major Issue 4.4: No statistical comparison with data.

Response: A χ^2 fit to seven $f\sigma_8(z)$ measurements has been added (Table 4). The best-fit model ($\alpha = -0.003$) achieves $\chi^2 = 4.49$ vs. ΛCDM ’s $\chi^2 = 6.08$, a mild improvement of $\Delta\chi^2 = -1.6$. We acknowledge that this is not statistically decisive ($< 1.4\sigma$ preference) and that a full MCMC analysis against Planck+BAO+SNe is required.

Major Issue 4.5: The vacuum phase transition is underconstrained.

The reviewer noted that the phase transition adds three parameters and that the nonlinear σ_8 suppression is unquantified. *Response*: The phase transition scenario is now explicitly labelled as **speculative** and separated from the main constant- α results. Numerical investigation revealed that, with vacuum energy properly sequestered from the Friedmann equation, the phase transition cannot simultaneously resolve both tensions within linear theory. The scenario requires nonlinear (N-body) treatment and is identified as a priority for future work, not a result of this paper.

Major Issue 4.6: The $\rho_{\text{vac}} \sim 4.5 \times 10^{-24} \text{ kg m}^{-3}$ value.

Response: A brief derivation is now included in Section 4.3: the value follows from equating the vacuum mass in a halo ($\frac{4}{3}\pi R^3 \rho_{\text{vac}}$) to the inferred dark matter mass, for a galaxy with $M_{\text{DM}} \sim 2.5 \times 10^{11} M_{\odot}$ and $R \sim 100$ kpc. We note that this value is model-dependent and explore the sensitivity of results to its uncertainty.

Minor Issues.

5.1 (Self-referencing): Key derivations from companion papers have been reproduced within the manuscript where essential for the argument.

5.2 (Water analogy): The analogy has been removed and replaced by the de Sitter–Schwarzschild analysis.

5.3 (Equation 4 domain): The equation has been withdrawn; see Issue 4.2 response.

5.4 (Table 1 labels): $\alpha = +0.01$ now correctly labelled as “worsens S_8 significantly,” not “mild.”

5.5 (Void fraction): The parametric form $f_v(z) = 0.6/[1 + \exp((z - 2)/1.5)]$ is motivated by void statistics from [Pan et al. \(2012\)](#) and [Hamaus et al. \(2014\)](#). Sensitivity analysis shows that varying the present-day f_v between 0.5 and 0.7, or the midpoint between $z = 1.5$ and $z = 2.5$, changes S_8 by $< 0.5\%$.

5.6 (Code validation): The code reproduces the known Λ CDM growth factor to machine precision ($|D_{\text{code}} - D_{\text{analytic}}|/D < 10^{-9}$ at $z = 0$, verified against the analytic matter-dominated solution $D \propto a$).

5.7 (Proposition 4.4 untested): The claim has been softened to “a qualitative prediction that can be tested with future data.”

5.8 (Section 10): The standalone journal recommendation section has been removed; a brief note is retained in the final paragraph.

Reviewer Questions.

Q1 (Matter-to-DE transition): The Friedmann equation is now clarified as **identical to Λ CDM** (Eq. 25). Vacuum energy is sequestered from the background; only the Poisson equation is modified. Late-time acceleration is driven by the geometric Λ_{geom} , not by vacuum energy.

Q2 (H_0 prediction): The model predicts the same H_0 as Λ CDM at the background level (sequestered vacuum). Any H_0 modification would require non-sequestered vacuum contributions, which are not part of the present framework.

Q3 (Israel junction wall density): The constraint yields $\sigma \sim \rho_{\text{vac}} R_v / c^2$, which for $\rho_{\text{vac}} \sim 10^{-24} \text{ kg m}^{-3}$ and $R_v \sim 30 \text{ Mpc}$ gives $\sigma \sim 10^8 M_{\odot}/\text{Mpc}^2$, consistent with observed void wall surface densities.

Q4 (α discrepancy with RVM): The RVM fits constrain the *temporally averaged* parameter $\nu \sim 10^{-3}$, giving $\alpha \sim 3 \times 10^{-4}$. The best-fit value in this paper, $\alpha = -0.003$, is consistent with this (within one order of magnitude). The values $\alpha \sim 0.01$ – 0.05 used in Tables 1–2 are *illustrative* upper bounds; the observationally preferred value is $\alpha = -0.003$.

Q5 (CMB quantitative check): With vacuum sequestered from Friedmann, the background $H(z)$ is identical to Λ CDM. The sound horizon, angular diameter distance, and CMB peak positions are preserved exactly. Only the ISW contribution (sub-percent) is modified.

Second Round

Placeholder: to be completed upon receipt of second-round reviews.

Further Clarifications

Placeholder: to be completed as needed.